



NATIONAL JUNIOR COLLEGE
SENIOR HIGH 2 PRELIMINARY EXAMINATION
Higher 2

NAME

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SUBJECT
CLASS

2ma2

REGISTRATION
NUMBER

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MATHEMATICS

9758/02

Paper 2

21 September 2020

3 hours

Candidates answer on the Question Paper.

Additional Materials: List of Formulae (MF26)
Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your name, class and registration number on the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use paper clips, glue or correction fluid.

Answer **all** the questions.

Write your answers in the spaces provided in the Question Paper.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use an approved graphing calculator.

Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers. Up to 2 marks may be deducted for improper presentation.

The number of marks is given in brackets [] at the end of each question or part question.

Question Number	Marks Possible	Marks Obtained
1	6	
2	7	
3	8	
4	8	
5	11	
6	7	
7	8	
8	9	
9	10	
10	11	
11	2	
12	13	
Presentation Deduction		– 1 / – 2
TOTAL	100	

This document consists of **27** printed pages and **1** blank page.

Section A: Pure Mathematics [40 marks]

- 1 (i) Given that $y = \sin^{-1}(2x) + 1$, by using standard series from the List of Formulae (MF26), show that $\frac{dy}{dx} \approx 2 + 4x^2 + 12x^4 + 40x^6$, and state the range of values of x for which the expression is valid. [4]

- (ii) Hence, find an approximate expression for y , up to and including the term in x^7 . [2]

- 2 Referred to the origin O , points A , B and C have position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively. B and C are points on a circle with diameter OA and circumference π units.

- (i) Give a geometrical interpretation of $|\mathbf{a} \cdot \mathbf{b}|$ and show that $\mathbf{a} \cdot \mathbf{b} = |\mathbf{b}|^2$. [3]

The point D is on OA such that $OD : DA = 4 : 5$. It is given that BD is perpendicular to OA .

- (ii) Find the cosine of the acute angle BOD . [2]

- (iii) Given that $|\mathbf{a} \times \mathbf{c}| = 0.4$, find the area of triangle OCD . [2]

- 3 In the controlled environment of Laboratory (Lab) A where cells are cultivated, the number of cells increases by 250% from the start of the week to the end of the week. At the end of each week, 1.2 million cells are transferred to another laboratory for experiments. The number of cells (in millions) in Lab A at the start of the n th week is given by a_n .

- (i) Show that $a_n = 3.5^{n-1}a_1 - 0.48(3.5^{n-1} - 1)$. [3]

It is given that Lab A begins operating at the start of the first week with 0.5 million cells.

- (ii) Find the number of cells in Lab A at the end of the third week, before the cells are transferred to another laboratory. [2]

A research requires at least 60 million cells before it can commence. To facilitate this, Lab A and Lab B are to cultivate and combine all the cells in both the labs by the start of a certain week. Lab B begins operating one week after Lab A . Lab B begins operating at the start of that week with 0.5 million cells and cultivates 5 million cells by the end of that week. At the end of each subsequent week, Lab B cultivates 5 million cells more than in the previous week.

- (iii) In which week can the research first commence? [3]

- 4 The curve C_1 has parametric equations

$$x = 4 + 4 \sin \theta, \quad y = 2 \cos \theta,$$

where $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$.

Sketch C_1 , labelling any points where C_1 crosses the axes.

[2]

The curve C_2 has equation $y = x^2 - ax$, where a is a positive constant.

- (a) In the case where $a = \frac{7}{2}$, it is given that $(4, 2)$ is a point on both C_1 and C_2 .

Sketch C_2 on the same diagram as C_1 , labelling the points of intersection of C_1 and C_2 .

Hence, find the exact area of the region bounded by C_1 and C_2 .

[5]

- (b) State the range of values of a such that C_1 and C_2 meet at two distinct points.

[1]

- 5 The complex number z is given by $z = re^{i\theta}$, where $r > 1$ and $-\pi < \theta < -\frac{1}{2}\pi$.

The complex number w is such that $|w^*z| = r^2$ and $\arg\left(-\frac{w}{z^3}\right) = \pi - 4\theta$.

- (i) Find $|w|$ in terms of r and show that $\arg w = -\theta$.

[4]

- (ii) State $|z + w|$ and indicate, on an Argand diagram, the points A , B , C and D that represent the complex numbers z , w , $z + w$ and zw respectively.

[5]

- (iii) Hence show that the area of the quadrilateral $ACBD$ is given by $r^2 \sin 2\theta - r^3 \sin \theta$.

[2]

Section B: Probability and Statistics [60 marks]

- 6 A machine has previously been considered capable of producing spherical pellets with mean diameters 12.00 mm. After some modification to the machine, a random sample of 105 spherical pellets was taken and the diameter, x mm, of each pellet is measured. The results obtained are summarised as follows.

$$\sum x = 1261.05 \quad \sum x^2 = 15145.47$$

- (i) Calculate unbiased estimates of the population mean and variance of the diameter of spherical pellets after modification. [2]
- (ii) Test, at the 3% level of significance, whether the mean diameter of the spherical pellets has increased after modification. [3]
- (iii) State the meaning of the p -value of the test in part (ii) in context. [1]
- (iv) Explain why there is no need for the production manager to know anything about the population distribution of the diameters of the spherical pellets. [1]

- 7 Independent events A and B are such that $P(A) = 0.45$ and $P(B) = b$.

- (i) Prove that A' and B are independent events. [2]

For a third event C , it is given that A and C are mutually exclusive, $P(B \cap C) = 0.2$ and $P(A' \cap B' \cap C') = 0.1$.

- (ii) Write down expressions for $P(A \cap B')$ and $P(A' \cap B \cap C')$ in terms of b . [2]
- (iii) Find exactly the maximum and minimum possible values of $P(A \cap B)$. [4]

- 8 Allysa has 20 blocks, consisting of 4 sets of 5 blocks. Each set consists of 1 triangle, 1 square, 1 parallelogram and 2 identical pentagons of the same colour. The colours of the sets are red, yellow, blue and green. So, for example, the blue set consists of 1 blue triangle, 1 blue square, 1 blue parallelogram and 2 blue pentagons.

- (a) Suppose Allysa picks 4 blocks and arranges them in a row. In how many different ways can this be done so that the 4 blocks include all colours and all shapes? [2]
- (b) Suppose instead Allysa chooses 5 blocks. Find the number of ways in which the 5 blocks can be chosen if they include exactly 2 pentagons, at least 1 triangle, and at least 1 square. [3]
- (c) All the blocks are now arranged in a circle. Find the number of different arrangements that can be made with all the pentagons together and no two triangles adjacent. [4]

- 9** In a game, a dealer tosses three fair six-sided dice, each numbered from 1 to 6 and a player pays a stake of \$ k to guess the numbers that appear on the top sides of the dice. The number of correct guesses the player makes in one game is denoted by X .

(i) Show that $P(X = 2) = \frac{5}{72}$ and tabulate the probability distribution of X . [3]

(ii) A sample of 30 observations of X is taken. Find the probability that the mean of this sample exceeds 1. [4]

For each game, the dealer pays back the stake if the player makes one correct guess, twice the stake if the player makes two correct guesses and ten times the stake if the player makes three correct guesses. The player loses the stake if there are no correct guesses.

(iii) If 3 games are played, find the probability that the player makes a net loss. [3]

- 10** (a) In a barber shop, the number of male customers who dye their hair in a particular day is denoted by X . It is given that X has the distribution $B(6, p)$.

(i) Given that $\text{Var}(X) = 0.8466$ and $p < 0.5$, find the value of p . [1]

(ii) Given instead that the probability that not all the male customers dye their hair is $\frac{117648}{117649}$, find the exact value of p . [2]

- (b) Analysis of the customer demographics of a hair salon over a long period shows that 65% of the customers are female. In a day, the first n customers that the salon receives are observed and the number of those customers who are female is denoted by Y .

(i) State, in context, two assumptions needed for Y to be well modelled by a binomial distribution. [2]

Assume now that Y has a binomial distribution.

(ii) Find the probability that, on a randomly chosen day, out of the first 15 customers, the thirteenth customer is the eighth and last female customer. [3]

(iii) Find the probability that, in seven randomly chosen days, there are exactly three days with at least 8 female customers among the first 15 customers. [3]

- 11** Research is being carried out into how the concentration of a drug in the bloodstream varies with time, measured from when the drug is given. Observations at successive times give the data shown in the following table.

Time (t minutes)	15	30	60	90	120	150	180	240	300
Concentration (x micrograms per litre)	82	65	43	37	22	19	12	6	2

Calculate the equation of the regression line of x on t . [2]

12 (a)

Weight Indicator Age (years)	Underweight	Acceptable Weight	Overweight
	< 5 th percentile	5 th - < 90 th percentiles	≥ 90 th percentile
13	≤ 15.2	15.2 – 25.0	≥ 25.0
14	≤ 15.5	15.5 – 25.5	≥ 25.5
15	≤ 15.9	15.9 – 26.1	≥ 26.1
16	≤ 16.2	16.2 – 26.5	≥ 26.5
17	≤ 16.4	16.4 – 27.0	≥ 27.0
18	≤ 16.7	16.7 – 27.4	≥ 27.4

The table contains information extracted from the health booklets for all new born babies in Singapore. It shows the BMI-for-age for boys aged 13-18 years in Singapore.

- (i) Assuming that the BMI for 18-year-old boys in Singapore is normally distributed with mean μ and variance σ^2 , use suitable information from the above chart to find the value of μ and show that $\sigma = 3.656$ correct to 3 decimal places. [4]
- (ii) State, with a reason, whether the BMI of the combined group of boys aged 13 and 18 years can be assumed to be normally distributed. [1]
- (b) During the circuit breaker period, a restaurant in a neighbourhood offers free delivery service to residents staying at Redwood Heights. On a randomly chosen day, the amount of time required for food preparation upon the placement of an order is normally distributed with mean 20 minutes and standard deviation 4 minutes. When the food is ready, a deliveryman collects the food immediately and delivers it on bicycle to Redwood Heights. The delivery time on bicycle from the restaurant to Redwood Heights is normally distributed with mean 18.1 minutes and standard deviation 2 minutes.
- (i) Find the latest time a food order has to be placed so that there is a 95% chance that the food is delivered to Redwood Heights by 7.00 pm. [4]

The restaurant owner also engages deliverymen who deliver on motorbikes. The delivery time on motorbike from the restaurant to Redwood Heights is normally distributed with mean 15 minutes and standard deviation 4 minutes. Both types of deliverymen are paid only for their delivery trips. Deliverymen by bicycles and by motorbikes are paid at a rate of \$1.00/min and \$1.20/min respectively.

- (ii) Find the probability that a deliveryman making 10 deliveries by motorbike gets paid more than a deliveryman making 10 deliveries by bicycle.
State, with a reason, which type of deliverymen would the cost-conscious restaurant owner employ. [4]